

18/09/2026.

Ex:- Poisson distⁿ, $P_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} = P(X=x)$

Let $X_1, X_2, \dots, X_n \sim P(\lambda)$

$$E(X_1) = E(X_2) = \dots = E(X_n) = \lambda, \quad E(\bar{X}) = \lambda$$

Now, $g(\lambda) = e^{-\lambda}$.

We want to estimate $g(\lambda)$.

Define a statistic:- $T(X_1) = \begin{cases} 1, & X_1 = 0 \\ 0, & X_1 \neq 0 \end{cases}$

$$E(T(X_1)) = 1 \cdot P(X_1 = 0) + 0 \cdot P(X_1 \neq 0) = 1 \cdot e^{-\lambda} = e^{-\lambda}$$

$\Rightarrow T(X_1)$ is unbiased estimate of $e^{-\lambda}$.

$$T(X_2) = \begin{cases} 1, & X_2 = 0 \\ 0, & X_2 \neq 0 \end{cases}$$

$$E(T(X_2)) = e^{-\lambda}$$

$\Rightarrow T(X_2)$ is unbiased estimate of $e^{-\lambda}$.

Unbiased estimator if it exist, it may not be reasonable.

$X \sim P(\lambda)$. find unbiased estimator of $e^{-3\lambda}$.

$$E((-2)^X) = \sum_{x=0}^{\infty} \frac{(-2)^x e^{-\lambda} \cdot \lambda^x}{x!} = e^{-3\lambda}$$

$\Rightarrow (-2)^X$ is unbiased estimator of $e^{-3\lambda}$.

But $(-2)^X$ is not a good estimator for $e^{-3\lambda}$. Because,

$e^{-3\lambda}$ is always positive but our estimator $(-2)^X$, $X=0,1,\dots$ will take values like 1, -2, 4, -8, ...

Note:- We have observed that unbiased estimator may exist, may not exist, if it exist it may not be reasonable in practice.

Ques:- How to deal these challenges?

Consistent Estimator:- An estimator $T_n = T(\underline{x})$ is said to be consistent for $g(\theta)$ if for any $\varepsilon > 0$

$$P(|T_n - g(\theta)| > \varepsilon) \longrightarrow 0 \text{ as } n \longrightarrow \infty$$

$\Rightarrow T_n$ converges to $g(\theta)$ in probability.

Convergence in probability:-

Let X_n be a seqⁿ of RVs and X be another random variable. We say that X_n converges to X in probability if for every $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0$$

$$|X_n - X| < \varepsilon$$

$$X_n \xrightarrow{P} X$$

$$\frac{1}{n}, n \in \mathbb{N}$$

$$\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$$

$$\rightarrow 0 \text{ as } n \rightarrow \infty$$

for given $\varepsilon > 0$, $|\frac{1}{n} - 0| < \varepsilon$

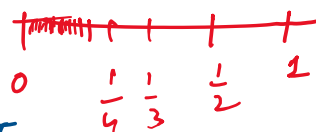
$$a_n \rightarrow a$$

for given $\varepsilon > 0$,

$$|a_n - a| < \varepsilon$$

Ex:- Let $X_1, X_2, \dots, X_n \sim U(0, \theta)$, $\theta > 0$.

We want to estimate θ using statistic $T_n = X_{(n)}$.



Ques:- Is T_n a consistent estimator for θ ?

Ans:- $P(|T_n - \theta| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$

$$P(X_{(n)} - \theta > \varepsilon) \text{ or } P(X_{(n)} - \theta < -\varepsilon)$$

$\Downarrow$

$$P(X_{(n)} > \theta + \varepsilon)$$

$\Downarrow$

$$P(X_{(n)} < \theta - \varepsilon)$$

CDF of $X_{(n)}$

$$= (F_X(n))^n$$

$U(0, \theta)$

$$F_X(n) = \frac{x - 0}{\theta - 0}$$

$$= P(X \leq x)$$

$$X_i \sim U(0, \theta), 0 \leq X_i \leq \theta$$

$$X_{(n)} \leq \theta$$

$$P(X_{(n)} > \theta + \varepsilon) = 0 \quad \because \theta - \varepsilon < \theta$$

$$= \left(\frac{(\theta - \varepsilon) - 0}{\theta - 0} \right)^n$$

$$\Rightarrow \left(\frac{\theta - \varepsilon}{\theta} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow P(X_{(n)} < \theta - \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$P(|T_n - \theta| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\Rightarrow T_n = X_{(n)}$ is consistent estimator of θ for $U(0, \theta)$.

Thm:- If $E(T_n) = \theta_n \rightarrow \theta$ and $\text{Var}(T_n) = \sigma_n^2 \rightarrow 0$ as $n \rightarrow \infty$.

Then T_n is consistent for θ .

Proof:-

$$P(|T_n - \theta| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$T_n - \theta = T_n - \theta_n + \theta_n - \theta$$

$$|T_n - \theta| \leq |T_n - \theta_n| + |\theta_n - \theta|$$

$$P(|T_n - \theta| > \varepsilon) \leq P(|T_n - \theta_n| + |\theta_n - \theta| > \varepsilon)$$

$$= P(|T_n - \theta_n| > \varepsilon - |\theta_n - \theta|)$$

$$P(|Y - E(Y)| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2} \quad , \quad \sigma^2 = \text{Var}(Y), Y \text{ RV}$$

$$P(|T_n - \theta| > \varepsilon) \leq \frac{\sigma_n^2}{(\varepsilon - |\theta_n - \theta|)^2} \longrightarrow 0 \text{ as } n \rightarrow \infty$$

$\Rightarrow T_n$ is consistent estimator for θ .

Ex:- Let $X_1, X_2, \dots, X_n$ be iid RV with $E(X_i) = \mu$ and $\text{var}(X_i) = \sigma^2$

Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

$$E(\bar{X}) = \mu, \text{var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\Rightarrow \bar{X}$ is consistent estimator for μ .

Ex:- $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$
 $(n-1) \frac{s^2}{\sigma^2} \sim \chi_{n-1}^2$

Verify that s^2 is consistent estimator for σ^2 .

Ans:- $E(s^2) = \sigma^2 \rightarrow \text{unbiased}$

$$\text{var}\left(\frac{(n-1)s^2}{\sigma^2}\right) = 2(n-1)$$

$$\text{var}(ax) = a^2 \text{var}(x)$$

$$\Rightarrow \frac{(n-1)^2}{\sigma^4} \text{var}(s^2) = 2(n-1)$$

$$\Rightarrow \text{var}(s^2) = \frac{\sigma^4 \cdot 2(n-1)}{(n-1)^2} = \frac{2\sigma^4}{n-1} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\Rightarrow s^2$ is consistent estimator for σ^2 .

Note:- In Poisson example, there can be many unbiased estimator for a given problem. So, how to decide which estimator is good.

Mostly, Mean Square Error (MSE) criterion is used to choose the better estimator.

19/09/2026.

MSE of an estimator:- Let $T(\underline{x})$ be an estimator of $g(\theta)$, then MSE of $T(\underline{x})$ is defined as

$$\begin{aligned} \text{MSE}(T(\underline{x})) &= E[T(\underline{x}) - g(\theta)]^2 \\ &= E\left[\underbrace{T(\underline{x}) - E(T(\underline{x}))}_A + \underbrace{E(T(\underline{x})) - g(\theta)}_B\right]^2 \\ &= E\left[\underbrace{(T(\underline{x}) - E(T(\underline{x})))^2}_{\text{var}(T(\underline{x}))} + E(T(\underline{x}) - g(\theta))^2 + 2 E(T(\underline{x}) - E(T(\underline{x}))) (E(T(\underline{x})) - g(\theta))\right] \\ &= \text{var}(T(\underline{x})) + \text{Bias}(T(\underline{x}))^2 \end{aligned}$$

Thus, if $T_1(\underline{x})$ and $T_2(\underline{x})$ be two estimators of $g(\theta)$, then T_1 is s.t.b better than T_2 if

$$\text{MSE}(T_1) \leq \text{MSE}(T_2) \quad \forall \theta \in \Theta$$

In particular, if $T(\underline{x})$ is unbiased for $g(\theta)$, then

$$\text{Bias}(T(\underline{x})) = 0$$

$$\text{So, } \text{MSE}(T(\underline{x})) = \text{var}(T(\underline{x}))$$

Now, if $T_1(X)$ and $T_2(X)$ are two unbiased estimators of $g(\theta)$, then T_1 is s.t.b better than T_2 if

$$\text{var}(T_1(X)) \leq \text{var}(T_2(X)) \quad \forall \theta \in \Theta$$

Note:- Goodness of estimator is decided by the smaller variance (in unbiased case).

Ex:- $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} P(\lambda)$

$$T_1 = X_1, \quad T_2 = \frac{X_1 + X_2}{2}, \quad T_3 = \bar{X}$$

To estimate λ ,

$$E(T_1) = \lambda, \quad E(T_2) = \lambda, \quad E(T_3) = \lambda$$

$$\text{var}(T_1) = \lambda$$

$$\text{var}(T_2) = \lambda/2$$

$$\text{var}(T_3) = \lambda/n$$

$\Rightarrow T_3$ is better than T_1 and T_2 if $n > 2$

Note:- If we compare two unbiased estimators of $g(\theta)$, then estimator with minimum variance is preferable.

Uniformly Minimum Variance Unbiased Estimator (UMVUE):-

An estimator $T(X)$ of $g(\theta)$ is said to be UMVUE of $g(\theta)$ if $T(X)$ is unbiased for $g(\theta)$ and for every other unbiased estimator $T^*(X)$ of $g(\theta)$, we have

$$\text{var}(T(X)) \leq \text{var}(T^*(X)) \quad \forall \theta \in \Theta$$

Note:- If $T(\underline{x})$ is UMVUE of $g(\theta)$, then $T(\underline{x})$ is unique.

Ex:- Let $X_1, X_2, \dots, X_n$ be a random sample from a popⁿ with mean μ and variance σ^2 .

$$T_1 = \bar{X}, \quad T_2 = \frac{2}{n(n+1)} \sum_{i=1}^n i X_i$$

Claim:- T_1 and T_2 are unbiased and consistent.

Ques:- How to evaluate UMVUE for a given estimation problem?

Method of lower bound
(Frochet Cramer Rao lower bound)

$\Downarrow$

Lao Blackwell Lehmann
Scheffe approach
involves

sufficiency completeness

Cramer Rao Inequality:- Let $X_1, X_2, \dots, X_n$ be a random sample with prob. distⁿ $f(x, \theta)$, $\theta \in \Theta$ and suppose certain regularity conditions are satisfied. Then for any unbiased estimator $T(\underline{x})$ of $\eta(\theta)$, a function of parameter θ , we have

$$\text{var}(T(\underline{x})) \geq \frac{(\eta'(\theta))^2}{n I(\theta)} \quad \forall \theta \in \Theta \quad \text{--- (1)}$$

where $I(\theta) = E \left[\frac{\partial}{\partial \theta} \log f(x, \theta) \right]^2$

The quantity $I(\theta)$ is known as Fisher information for θ in a single RV X .

Regularity Conditions:-

(1) θ lies in an open interval Θ of the real line.

$N(\mu, \sigma^2)$

$P(\lambda)$

$U(0, \theta)$

$(\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$

$\lambda \in \mathbb{R}$

$\theta \in \mathbb{R}$

(2) $\frac{\partial f(x, \theta)}{\partial \theta}$ exists $\forall \theta \in \Theta \forall x$.

(3) The range of integration or summation is independent of parameter θ .

(4) The integrals or sums are uniformly convergent.

(5) $I(\theta) = E \left(\frac{\partial}{\partial \theta} \log f(x, \theta) \right)^2$ exists and is positive for all θ .

Particular case:- If $T(X)$ is unbiased estimator of parameter θ , then

$$E(T(X)) = \theta \quad \text{i.e.} \quad \eta(\theta) = \theta \quad \text{i.e.} \quad \eta'(\theta) = 1$$

$$\Rightarrow \text{Var}(T(X)) \geq \frac{1}{n I(\theta)} \quad \text{--- (2)}$$

Note:- $I(\theta)$ can also be expressed as

$$I(\theta) = -E \left[\frac{\partial^2}{\partial \theta^2} \log f(x, \theta) \right]$$

We can apply CR inequality to obtain UMVUE for $\theta / \eta(\theta)$. The following steps are useful in this regard.

- (1) Calculate $I(\theta)$ i.e. Fisher information.
- (2) Find CR lower bound appearing in (1)/(2).
- (3) Try to identify an unbiased estimator whose variance is equal to the CR lower bound.
- (4) If such an estimator is found out, then declare that estimator as the UMVUE of θ .

Ex:- Let $X \sim \text{Bin}(m, p)$, m -known, p -unknown
Find UMVUE of p .

Ans:- $f_X(x, p) = \binom{m}{x} p^x (1-p)^{m-x}$, $x = 0, 1, \dots, m$

$$\log f_X(x, p) = \log \binom{m}{x} + x \log p + (m-x) \log(1-p)$$

$$\frac{\partial \log f(x, p)}{\partial p} = 0 + \frac{x}{p} - \frac{m-x}{1-p} = \frac{x - mp}{p(1-p)}$$

$$E \left[\frac{\partial \log f(x, p)}{\partial p} \right]^2 = E \left[\frac{x - mp}{p(1-p)} \right]^2 = \frac{mp(1-p)}{p^2(1-p)^2}$$

$$I(\theta) = \frac{m}{p(1-p)}$$

By (1), $\text{var}(T(\underline{x})) \geq \frac{p(1-p)}{m}$

Guess an unbiased estimator of p .

Note that $E\left(\frac{X}{m}\right) = p$

$$\text{var}\left(\frac{X}{m}\right) = \frac{1}{m^2} \text{var}(X) = \frac{mp(1-p)}{m^2} = \frac{p(1-p)}{m}$$

$\Rightarrow$ variance of estimator X matches with CR lower bound.

So, $T(\underline{x}) = \frac{X}{m}$ is UMVUE of p .

Q1 Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} P(\lambda)$. Find UMVUE of λ . ($\bar{X}$)

Q2i Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

(i) Find UMVUE of μ when σ^2 is known. ($\bar{X}$)

(ii) Find UMVUE of σ^2 when μ is known ($\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$)

Q3:- Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\frac{1}{\theta})$. Find UMVUE of θ . ($\bar{X}$)

$$f(x, \theta) = \begin{cases} \frac{1}{\theta} \exp(-\frac{x}{\theta}), & 0 < x < \infty \\ 0, & \text{else} \end{cases}$$

$$\log f(x, \theta) = -\log \theta - \frac{x}{\theta}$$

$$E(X) = \theta$$
$$\text{Var}(X) = \theta^2$$

$$\frac{\partial}{\partial \theta} \log f(x, \theta) = -\frac{1}{\theta} + \frac{x}{\theta^2}$$

$$\Rightarrow E\left[\frac{\partial}{\partial \theta} \log f(x, \theta)\right]^2 = E\left[-\frac{1}{\theta} + \frac{x}{\theta^2}\right]^2$$

$$= E\left[\frac{x^2}{\theta^4} + \frac{1}{\theta^2} - \frac{2x}{\theta^3}\right]$$

$$= \frac{2\theta^2}{\theta^4} + \frac{1}{\theta^2} - \frac{2\theta}{\theta^3} = \frac{1}{\theta^2}$$

$$\Rightarrow I(\theta) = \theta^2$$

$$\Rightarrow \text{Var}(T(\underline{X})) \geq \frac{\theta^2}{n}$$

Guess an unbiased estimator $T(\underline{X}) = \bar{X}$

$$E(\bar{X}) = E\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{n\theta}{n} = \theta$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{\sum X_i}{n}\right) = \frac{1}{n^2} \text{Var}(\sum X_i) = \frac{n\theta^2}{n^2} = \frac{\theta^2}{n}$$

$\Rightarrow \text{Var}(\bar{X})$ coincides with CR lower bound.

$\Rightarrow \bar{X}$ is UMVUE of θ .